SIMShield

Network-native identity verification & continuous KYC fraud prevention

GSMA Open Gateway APIs · Nokia NaC Platform · Google Gemini AI

Open Gateway Hackathon 2026 · Talent Arena

SIMShield

Payment fraud prevention
powered by Nokia Network as Code

Open Gateway Hackathon 2026 · Talent Arena · MWC

Nokia NaC · GSMA CAMARA · Movistar · Orange · Vodafone

Every fraud attack starts with a SIM card nobody checked.

SIMShield turns the mobile network into a real-time fraud detector — before every transaction completes.

- SIM Swap
- KYC Match
- Device Swap
- Device Location

< 300ms	5 APIs	3 MNOs
verification	Nokia NaC	~95% Spain

sim-shield.vercel.app

THE PROBLEM

Fraudulent transactions pass every standard check. The SIM is never verified.

SIMShield

THE SCENARIO

María is shopping in Madrid. 450km away in Seville, a fraudster tries to use her cloned card at a POS terminal.

The bank fires a verification request to SIMShield. We check via Device Location API: SIM swapped 38 minutes ago. Device detected in Seville. Card registered in Madrid.

SIMShield's chatbot alerts María: "Are you currently in Seville?" · She replies: No. · Fraud stopped.

Score

9 / 100

DENY

Scale — higher is safer: < 40 — Deny · 40–69 — Clarify via chatbot · ≥ 70 — Approve

£300M

M&S fraud loss

SIM swap via helpdesk · Q1 2025

1×

SIM swap attacks

Rising sharply · Spain following UK curve

€2.1B

Card-present EU fraud

POS terminal attacks · phone ≠ card location

WHY CURRENT DEFENCES FAIL

SMS 2FA

OTP routes to attacker once swap completes. Bank has zero visibility.

Bank fraud AI

Fires after transaction. SIM swap happened minutes before.

Static KYC

Checked once at onboarding. Never re-verified when SIM changes.

Carrier PINs

Helpdesk bypass under pressure. The exact M&S attack vector.

HOW IT WORKS

5 Nokia NaC APIs fire in parallel · Score 0–100 (higher = safer) · Decision in < 300ms

SIMShield

The bank calls our API. We check the network. We return a score. If the score is too low — the transaction does not complete.

01

POS trigger

Card used. Bank captures phone number, POS coordinates, transaction amount.

02

Verification request

Bank calls SIMShield silently. Customer sees nothing. No friction.

03

5 APIs fire

Location Retrieval · Location Verification · SIM Swap · Device Swap · KYC Match

04

Score computed

0 to 100. Higher = safer. Weighted by fraud prediction strength per signal.

05

Decision returned

≥ 70 approve · 40–69 cl deny

287 ms avg

SCORING SYSTEM

≥ 70 APPROVE

Phone near POS · SIM stable · KYC confirmed · Device unchanged

40–69 CLARIFY

Ambiguous signals · Telegram chatbot contacts cardholder · human confirms

< 40 DENY

SIM swapped · device 450km from POS · Score 9/100 · card block suggested

TECH STACK

Next.js + TypeScript	Python FastAPI	Nokia NaC SDK	Vercel	Telegram Bot	Gemini + MCP
Frontend	Backend	5 live APIs	Deployment	Clarify flow	AI agent

PRODUCT & API ARCHITECTURE

End-to-end flow · Nokia NaC integration · AI layer · Live on Vercel

SIMShield

USER LAYER

Merchant POS · Bank App · Enterprise Helpdesk · ADP / HR Platform

SIMSHIELD PRODUCT LAYER

Next.js frontend · Python FastAPI backend · Vercel · sim-shield.vercel.app

network-as-code.developer.nokia.com · GSMA CAMARA standard · Single SDK, multi-operator

Location Retrieval

LIVE

Get SIM location from network

Location Verification

LIVE

SIM vs POS distance check

SIM Swap

LIVE

Swapped in last 24 hours?

Device Swap

Hardware fingerprint changed?

AI LAYER

```
check_location()
check_sim_swap()
device_swap()
kyc_match()
```

GEMINI AGENT

Calls all 4 MCP tools in parallel. Reasons across signals. Returns score + plain-English verdict.

OPERATOR NETWORK

Movistar · Orange · Vodafone · 3 live 5G networks · ~95% Spanish mobile subscribers

GEMINI VERDICT — DEMO MOMENT

"SIM swapped 38 min ago · Device: Seville · Card registered: Madrid · KYC mismatch confirmed · Score: 9 / 100 · Recommendation: DENY"

DENIED
9 / 100

COMMERCIAL MODEL

Target clients · Phased financial model · ROI · Why now

SIMShield

TARGET CLIENT SEGMENTS

BEACHHEAD — MVP

Spanish Neobanks & Fintechs

Rebellion Pay · Bnext · Revolut ES

€0.004 / call

SCALE — PROCESSORS

Payment Processors

Redsys — 1 contract = national coverage

Rev share

ENTERPRISE MNC

ADP · Workday · Large Banks

Helpdesk impersonation = payroll fraud

€8 / seat / mo

SME

Spanish SMEs via Plugin

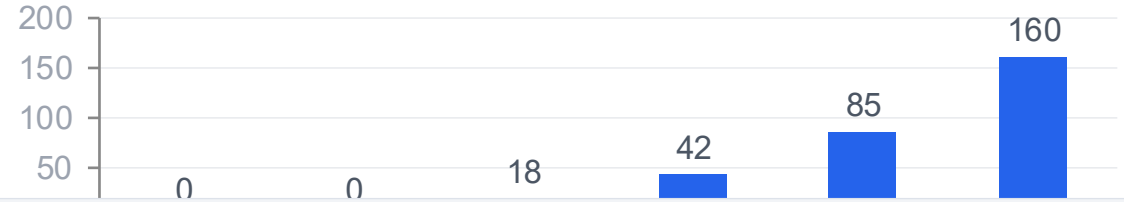
200K via Holded + Quipu

€29 / mo

ROI — 1 prevented fraud = > 116,000 checks paid for · > 2,000× return · ~€0.036 per check

PHASED FINANCIAL MODEL

	PHASE 1 MVP · Now → Q2 2026	PHASE 2 MVT · Q3 2026→Q1 2027	PHASE 3 Scale · Q2–Q4 2027
Clients	PoC only	20 fintechs + 2 MNC	
Checks / mo	Demo	~2M	
ARR	€0	~€85K	
Cost / check	€0.036	€0.036	
Break-even	—	Month 8–10	
MNO rev share	—	15%	
Spain API $$	Not live	Launching	



TAM **$47B** SAM **$8.4B** SOM **$220M**

Viable for transactions > €200 · Threshold logic skips micro-transactions · Germany live, Spain 2026

NOKIA BUILT THE PIPES. SIMSHIELD BUILT THE PRODUCT.

4 specific gaps filled · commercial product layer on Nokia NaC · scalable globally

SIMShield

NOKIA + TELEFÓNICA TESTBED VS SIMSHIELD

Dimension	Nokia + Telefónica (Lab)	SIMShield (Product)
APIs	SIM Swap + Device Swap (2)	Location · Verification · SIM Swap · Device Swap · KYC Match (5)
Operators	Telefónica only	Movistar + Orange + Vodafone (~95% Spain)
KYC signals	None	Live name / DOB match vs operator record in real time
Geolocation	None	Device Location vs POS coordinates — card-present fraud stopped
AI layer	Protocol testing	Gemini decisions via MCP · scored output · human-in-loop
Commercial	Lab PoC, no pricing	€0.004/call · €29 SME · €8/seat MNC — deployable today
Compliance	Not addressed	DORA-ready · zero-data broker · GDPR Art. 6(1)(f)

Nokia built the infrastructure. Telefónica proved the protocol. We built the commercial product nobody has built on top. Nokia NaC is a developer platform — they need application companies. SIMShield is that company.

SCALABILITY

- GSMA CAMARA → any operator globally
- Nokia NaC SDK → one integration, all MNOs
- ADP / Workday → 1 contract, all employees

WHY NOW

- DORA 2025 — real-time identity mandatory
- Spain API monetisation launching 2026
- M&S attack vector happening today at scale

OPERATOR ROI

- Lower chargebacks + regulatory pressure
- 15% rev-share aligns operator incentives
- Proactive fraud cut — not reactive enforcement

SIMShield

Network-native identity verification & continuous KYC fraud prevention

- **287ms** avg verification latency
- **€0.036** cost per check
- **> 2,000×** ROI per prevented fraud

The operators are in this room.

The regulation is already in force.

The only thing missing was the product.

That product is SIMShield

Contact Us

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